

Mg TRINITY DN1546 c0 g1 serine rm 1	TPGAKTDSGDGDVD.....SIVLPSTPKGSGPGF.	28
Mg TRINITY DN1546 c0 g1 serine rm 2	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPGF.	76
Mg TRINITY DN1546 c0 g1 serine rm 3	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPAF.	76
Mg TRINITY DN1546 c0 g1 serine rm 4	PGTTPGVVTGPDGQP.V.KYIVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVD.....SIVLPSTPKGSGPGF.	76
Mlaby 7048 AgSp1 serine rm1a 1	PGTTPGAETGPDGNVD.....KIILPTTPVRPEPQQ	32
Mlaby 7048 AgSp1 serine rm1a 3	PGTTPGVVTGPDGQP.V.QFIVPQGAFTTPGSIPGPNKPVH.VRPAGPGTTPGAKTDSGSIQ.....SIVLPSTPKPSGPGF.	76
Mlaby 7048 AgSp1 serine rm1a 12	FIVPQGAFTTPGSIPGPKGKPVH.VRPAGPGTTPGAKTDSGNID.....SIVLPSTPKPSRPGK.	59
Mlaby 7048 AgSp1 serine rm1a 14	FIVPQGAFTTPGSIPGPDGKQVH.VRPAGPGTTPGAK.....	36
Mlaby 7048 AgSp1 serine rm1a 25		0
Mlaby 7048 AgSp1 serine rm1a 26	PGTTPGVVTGPGGKPL.R.....	17
Mlaby 7048 AgSp1 serine rm1a 27		0
Mlaby 7048 AgSp1 serine rm1a 28	PGTTPGTVTGPDGNP.VKFIVPQGAFTTPGSISGPNKPIH.VRPALPGTTPGAK.....	53
Mlaby 7048 AgSp1 serine rm1a 37		0
Mlaby 7048 AgSp1 serine rm1a 38	PGTTPGVVTGPGGQPL.RFIVPQGAFTTPGSIPGPNKQPIH.VRPAGPGTTPGAKTDSGNID.....SIVLPSTPKGSGPVP.	76
Ncruc 58532and13532 AgSp1 serine rm 87	SPETSNG	7
Ncruc 58532and13532 AgSp1 serine rm 88	PGTTPGFVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VKPAGPGTTPGAKTDSGSD.....SIVLPSTPK.....	70
Ncruc 58532and13532 AgSp1 serine rm 89	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VKPAGPGTTPGAKTNSDGSID.....SIVLPSTPK.....	70
Ncruc 58532and13532 AgSp1 serine rm 90	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VRPAGPGTTPGAKTDSGSD.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 91	PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGTIKGPHGKPIH.VGPAGPGTTPGAKTDSGSD.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 92	PGTTPGVVTGPDGQP.VKFIVPQGAFTTPGTIPGPHGKPIH.IGPAGPGTTPGAKXDSGSD.....SIVLPSTPK.....	71
Ncruc 58532and13532 AgSp1 serine rm 93	PGTTPGVVTGPDGQP.VKFFVPNGAFSTPGSIPGPHGKPIH.IGPAGPGTTPGAKLDSGDID.....EILLPSTP.SP.....	71
Msag 19329 AgSp1.1 serine rm 1	AFTTPGSIAGPHGKRVH.VKPAGPELHQE.LKPSDGDVS.....SIVLPSTPKGSG.PG.	51
Msag 19329 AgSp1.1 serine rm 2	PGTTPGVVTGPDGQP.VKYVVPQ.EPYDSWIYCRTSRKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPKGSG.PG.	74
Msag 19329 AgSp1.1 serine rm 3	PGTTPGVVTGPDGQP.VKHVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPKASG.PG.	75
Msag 19329 AgSp1.1 serine rm 4	PGTTLAVVTGPDGQP.VKYVVPQVAFSTTPGSIAGPHGKRVHAREPAGPGTTPGAKTDSGDVS.....SIVLPSTPKGSG.PG.	76
Msag 19329 AgSp1.1 serine rm 5	PGTTPGVVTGPDGQSQFKYVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDPDGDVS.....SIVLPSTPKGSG.PG.	76
Msag 19329 AgSp1.1 serine rm 6	PGTTPGVVTGPDGQP.VKYVVPQGAFTTPGSIAGPHGKRVH.VKPAGPGTTPGAKTDSGDVS.....SIVLPSTPK.....	70
Lcorn 58650 AgSp1.2 serine rm1 3		0
Lcorn 58650 AgSp1.2 serine rm1 4	PGTTPGVVTGPDGQP.VKFIVPQGAFTTPGSIPGPNKPIH.VKPAGPGTTPGAKTNSDGSIDSIV.....	64
Lcorn 58650 AgSp1.2 serine rm1 5	PGTTPGVVTGADGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 6	FIVPQGAFTTSGSIPGPNKPIH.VKPAGPGTTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 7		0
Lcorn 58650 AgSp1.2 serine rm1 8	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 10		0
Lcorn 58650 AgSp1.2 serine rm1 11	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 12	FIVPQGAFTTPGSILGPNKPIH.VKPAGPGTTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 17	FIVPQGAFTTPGSIPGPNKPIR.....	23
Lcorn 58650 AgSp1.2 serine rm1 21		0
Lcorn 58650 AgSp1.2 serine rm1 22	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 45	TNSDGNID.....SIVLPSTPK.....	17
Lcorn 58650 AgSp1.2 serine rm1 56		6
Lcorn 58650 AgSp1.2 serine rm1 57	PGTSPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 69		0
Lcorn 58650 AgSp1.2 serine rm1 70	PGTSPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 75		0
Lcorn 58650 AgSp1.2 serine rm1 76		0
Lcorn 58650 AgSp1.2 serine rm1 77	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 79	IDSNGSID.....SIVLPSTPKGSGP...	21
Lcorn 58650 AgSp1.2 serine rm1 80	PGTTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 82		0
Lcorn 58650 AgSp1.2 serine rm1 83	PGTTPGVVTGPDGQP.VKFIVPNGAFSTPGSIPGPNKPIH.VGPAG.....	45
Aarg AgSp1 FC511 serine rm1 14	PGTTPGTMGSDGKPK.NKFVVPKGAFTTPGSFPGPDGKPIH.VEPAGPGSTPGTGTGPDGSIK.....QVFPVPTPKSSESSGG	77

Mg TRINITY DN1546 c0 g1 serine rm 1	QS.....	PTTPSHVTKPDGQPIQIVP	49
Mg TRINITY DN1546 c0 g1 serine rm 2	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mg TRINITY DN1546 c0 g1 serine rm 3	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mg TRINITY DN1546 c0 g1 serine rm 4	QS.....	PTTPSHVTKPDGQPIQIVP	97
Mlaby 7048 AgSp1 serine rm1a 1	N.....	TTTKVKGPQGKPLVVP	50
Mlaby 7048 AgSp1 serine rm1a 3	N.....	FQTPSPVTKPDGQPIQIVP	96
Mlaby 7048 AgSp1 serine rm1a 12	N.....	QFKNPPLPVTQSDGQPIQIVP	79
Mlaby 7048 AgSp1 serine rm1a 14	N.....		36
Mlaby 7048 AgSp1 serine rm1a 25	N.....	NPLPVTQSDGQPIQIVP	17
Mlaby 7048 AgSp1 serine rm1a 26	N.....		17
Mlaby 7048 AgSp1 serine rm1a 27	GPGGQPRPQSPG.....	QQTTPPEPTPGPMCKPLQIVP	33
Mlaby 7048 AgSp1 serine rm1a 28	N.....		53
Mlaby 7048 AgSp1 serine rm1a 37	N.....	NPLPVTQPDGQPIQIVP	17
Mlaby 7048 AgSp1 serine rm1a 38	ITTT.....	ITTTLPITPGPGPEPLQIVP	94
Ncruc 58532and13532 AgSp1 serine rm 87	K.....KGTGENPQNNANSLGPLQPDSTTTTPGSQPGNPMGPGSGPGGQTTTPGSQGPVDFD.....	HTPSPITTKPDGKPIQIVP	79
Ncruc 58532and13532 AgSp1 serine rm 88	GSGPGFQF.....	QTPQPVTTKPDGQPIQIVP	96
Ncruc 58532and13532 AgSp1 serine rm 89	GSGPGFQF.....	QTPQPVTTKPDGQPIQIVP	96
Ncruc 58532and13532 AgSp1 serine rm 90	SGPGFPPTPKGSGPGFQF.....	QTPQPVTTKPDGQPIQIVP	107
Ncruc 58532and13532 AgSp1 serine rm 91	SG.....PGFNF.....	ETPRPAPQSGGPIQIVP	95
Ncruc 58532and13532 AgSp1 serine rm 92	SG.....PGYNF.....	PTPRPAPQSGGPIQIVP	95
Ncruc 58532and13532 AgSp1 serine rm 93	PAP.....QPA.....	NPTTPTDVRGPQGSPIQIVP	97
Msag 19329 AgSp1.1 serine rm 1	F.....	QSTTPSQVTTKPDGQPIQIVP	72
Msag 19329 AgSp1.1 serine rm 2	F.....	HTPTPSQVTTKPDGQPIQIVP	95
Msag 19329 AgSp1.1 serine rm 3	F.....	QSTTPSQVTTKPDGQPIQIVP	96
Msag 19329 AgSp1.1 serine rm 4	F.....	RTPTLSQVTTKPDGQPIQIVP	97
Msag 19329 AgSp1.1 serine rm 5	F.....	QSTTPSQVTTKPDGQPIQIVP	97
Msag 19329 AgSp1.1 serine rm 6	GSGPG.....	FQFQTPQPITKPGGQPIQIVP	70
Lcorn 58650 AgSp1.2 serine rm1 3	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 4	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	96
Lcorn 58650 AgSp1.2 serine rm1 5	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 6	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	36
Lcorn 58650 AgSp1.2 serine rm1 7	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	10
Lcorn 58650 AgSp1.2 serine rm1 8	LPSTPKGSRPGFQF.....	QTPQPITKSGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 10	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 11	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 12	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	36
Lcorn 58650 AgSp1.2 serine rm1 17	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	23
Lcorn 58650 AgSp1.2 serine rm1 21	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	10
Lcorn 58650 AgSp1.2 serine rm1 22	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 45	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 56	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 57	GSGTG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 69	GSGTG.....	FQFQTPQPATKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 70	GSGTG.....	FQFQTPQPATKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 75	GSGTG.....	FQFQTPQPATKPGGQPIQIVP	11
Lcorn 58650 AgSp1.2 serine rm1 76	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 77	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 79	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	43
Lcorn 58650 AgSp1.2 serine rm1 80	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	17
Lcorn 58650 AgSp1.2 serine rm1 82	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	26
Lcorn 58650 AgSp1.2 serine rm1 83	ASGPG.....	FQFQTPQPITKPGGQPIQIVP	45
Aarg AgSp1 FC511 serine rm1 14	KKGTGENPANSNNAIGPLQT.....	TSTTTPTSISGPDGQPMQVVP	118

Mg TRINITY DN1546 c0 g1 serine rm 1	AG	51
Mg TRINITY DN1546 c0 g1 serine rm 2	AG	99
Mg TRINITY DN1546 c0 g1 serine rm 3	AG	99
Mg TRINITY DN1546 c0 g1 serine rm 4	AG	99
Mlaby 7048 AgSp1 serine rm1a 1	AG	52
Mlaby 7048 AgSp1 serine rm1a 3	AG	98
Mlaby 7048 AgSp1 serine rm1a 12	AG	81
Mlaby 7048 AgSp1 serine rm1a 14	..	36
Mlaby 7048 AgSp1 serine rm1a 25	AG	19
Mlaby 7048 AgSp1 serine rm1a 26	..	17
Mlaby 7048 AgSp1 serine rm1a 27	AG	35
Mlaby 7048 AgSp1 serine rm1a 28	..	53
Mlaby 7048 AgSp1 serine rm1a 37	AG	19
Mlaby 7048 AgSp1 serine rm1a 38	AG	96
Ncruc 58532and13532 AgSp1 serine rm 87	AG	81
Ncruc 58532and13532 AgSp1 serine rm 88	AG	98
Ncruc 58532and13532 AgSp1 serine rm 89	AG	98
Ncruc 58532and13532 AgSp1 serine rm 90	AG	109
Ncruc 58532and13532 AgSp1 serine rm 91	AG	97
Ncruc 58532and13532 AgSp1 serine rm 92	AG	97
Ncruc 58532and13532 AgSp1 serine rm 93	AG	99
Msag 19329 AgSp1.1 serine rm 1	AG	74
Msag 19329 AgSp1.1 serine rm 2	AG	97
Msag 19329 AgSp1.1 serine rm 3	AG	98
Msag 19329 AgSp1.1 serine rm 4	AG	99
Msag 19329 AgSp1.1 serine rm 5	AG	99
Msag 19329 AgSp1.1 serine rm 6	..	70
Lcorn 58650 AgSp1.2 serine rm1 3	AG	28
Lcorn 58650 AgSp1.2 serine rm1 4	AG	98
Lcorn 58650 AgSp1.2 serine rm1 5	..	17
Lcorn 58650 AgSp1.2 serine rm1 6	..	36
Lcorn 58650 AgSp1.2 serine rm1 7	AG	12
Lcorn 58650 AgSp1.2 serine rm1 8	..	17
Lcorn 58650 AgSp1.2 serine rm1 10	AG	28
Lcorn 58650 AgSp1.2 serine rm1 11	..	17
Lcorn 58650 AgSp1.2 serine rm1 12	..	36
Lcorn 58650 AgSp1.2 serine rm1 17	..	23
Lcorn 58650 AgSp1.2 serine rm1 21	AG	12
Lcorn 58650 AgSp1.2 serine rm1 22	..	17
Lcorn 58650 AgSp1.2 serine rm1 45	..	17
Lcorn 58650 AgSp1.2 serine rm1 56	AG	28
Lcorn 58650 AgSp1.2 serine rm1 57	..	17
Lcorn 58650 AgSp1.2 serine rm1 69	AG	28
Lcorn 58650 AgSp1.2 serine rm1 70	..	17
Lcorn 58650 AgSp1.2 serine rm1 75	..	11
Lcorn 58650 AgSp1.2 serine rm1 76	AG	28
Lcorn 58650 AgSp1.2 serine rm1 77	..	17
Lcorn 58650 AgSp1.2 serine rm1 79	AG	45
Lcorn 58650 AgSp1.2 serine rm1 80	..	17
Lcorn 58650 AgSp1.2 serine rm1 82	AG	28
Lcorn 58650 AgSp1.2 serine rm1 83	..	45
Aarg AgSp1 FC511 serine rm1 14	AG	120

 non-conserved
 similar
 $\geq 50\%$ conserved